

RESEARCH ARTICLE

FogCast: Probabilistic medium-range fog forecasting using global machine-learning weather prediction model

Prasad Deshpande^{1,2}  | Shubhangi Agarwal³ | Kritika Bansal¹ |
Arnab Bhattacharya³ | Shivam Tripathi¹

¹Department of Civil Engineering, Indian Institute of Technology, Kanpur, India

²Department of Agronomy, Kansas State University, Manhattan, Kansas, USA

³Department of Computer Science and Engineering, Indian Institute of Technology Kanpur, Kanpur, India

Correspondence

Prasad Deshpande, Department of Civil Engineering, Indian Institute of Technology Kanpur, India.

Email:

prasad.walchandsangli@gmail.com

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Abstract

This study presents FogCast, a Bayesian neural network (BNN)-based model that forecasts fog using the outputs of GraphCast, a machine-learning global weather prediction system. Compared to existing operational forecasts limited to 3–5 days lead time, FogCast extends fog forecasting capabilities to 10 days, making it a valuable tool for medium-range forecasts. FogCast provides probabilistic fog/no-fog forecasts while accounting for aleatoric and epistemic uncertainties, thereby improving forecast reliability. FogCast is trained and validated using visibility observations for seven cities in the Indo-Gangetic Plains from 2017 to 2023. The model achieves Critical Success Index (CSI) values between 0.44 (for six-hour lead time) and 0.36 (for 10-day lead time). Moreover, it correctly detects over 90% of very dense fog observations (visibility <50 m). The uncertainty analysis shows that aleatoric uncertainty is typically higher than epistemic uncertainty, suggesting potential avenues for forecast improvements. The real-time fog forecasts generated by FogCast are accessible at https://fog.iitk.ac.in/fog-prediction/fogcast_lite_forecast.php.

KEYWORDS

Bayesian neural networks, GraphCast, Indo-Gangetic Plains, mist, uncertainty, visibility

1 | INTRODUCTION

Fog is a prevalent and disruptive weather phenomenon at various places on Earth, including the Indo-Gangetic Plains (IGP). It significantly affects transportation systems, including air, rail and road, leading to delays, cancellations and an increased risk of accidents. These disruptions have considerable economic consequences — for instance, fog-related disturbances at New Delhi airport (India) alone resulted in losses of approximately 3.9 billion USD (~3000

Crore INR) between 2011 and 2016 (Kulkarni *et al.*, 2019). Beyond transportation, fog poses severe health risks, especially when combined with smoke, which creates smog, a hazardous air quality condition (Terry *et al.*, 2018). Additionally, foggy weather is associated with mental health issues, such as depression and gloom (Zheng *et al.*, 2019; Zheng *et al.*, 2024). From an agricultural perspective, fog impacts plant growth by reducing sunlight availability for photosynthesis, though it also provides dew water that can benefit crops (Gnanamoorthy *et al.*, 2022).

Despite its relatively low probability of occurrence, fog holds significant disaster potential in IGP, i.e., a region home to one-seventh of the world's population. Accurate

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and timely fog forecasting is thus crucial to mitigate its adverse impacts. Fog forecasting is also aligned with two United Nations Sustainable Development Goals, viz., 'Climate Action' and Sustainable Cities and Communities', and indirectly supports goals such as 'Decent Work and Economic Growth' and 'Good Health and Well-Being'.

Fog is defined as a cloud formed at ground level that consists of tiny water droplets suspended in the air, significantly reducing visibility to less than 1 km (WMO, 2011). Fog is a meteorological phenomenon that depends on complex interactions of several hydro-meteorological variables, including temperature, specific humidity, wind speed, and atmospheric pressure. Near-surface cooling, particularly during night-time, often leads to high relative humidity levels, facilitating fog formation when the air temperature approaches the dew point. Wind speed also plays a crucial role, as calm or low wind conditions reduce atmospheric mixing, enabling moisture to accumulate near the surface. Additionally, specific humidity and vertical velocity in the lower atmospheric layers, e.g., 1000 hPa, influence the availability of moisture and the stability of the atmosphere, both of which are crucial for fog development.

Fog is influenced by both local weather conditions and large-scale climate processes, making its forecast a challenging task. The complexities involved in fog formation have spurred numerous research efforts aimed at improving fog forecasting. Forecasting methods can be categorised in several ways: by lead time/forecast horizons (nowcasting, medium-range forecasting, and long-range/seasonal forecasting) and by classification type (binary and multiclass). Binary forecasts are fog or no fog, whereas multiclass forecasts are classes associated with different visibility ranges. Unlike classification, some forecasts provide the visibility value directly (i.e., regression). Lastly, fog forecasting approaches are divided into two categories based on methodology: statistical methods and physics-based methods.

Statistical (also known as data-driven) methods use past meteorological observations to identify patterns in fog occurrence. These methods range from classical time-series models (e.g., ARIMA; Srivastava *et al.*, 2017) to advanced machine-learning techniques like deep learning (e.g., LSTMs; Miao *et al.*, 2020). Statistical approaches are typically location-specific, relying on hydro-meteorological time-series from specific sites, and their ability to generalise to larger regions is limited. Despite their limitations, statistical methods are widely used due to their computational efficiency and simplicity (Arun *et al.*, 2022; Colabone *et al.*, 2015; Sharma *et al.*, 2024). Recent advancements in fog forecasting include the use of satellite-based observations in data-driven models (Bari *et al.*, 2023; Kim *et al.*, 2023).

Since these methods rely solely on satellite inputs without modelling atmospheric evolution, they are useful only for nowcasting (short-range forecasting with a lead time of up to 12 hours).

Physics-based methods rely on numerical weather prediction (NWP) models, which simulate primary atmospheric variables like temperature, wind speed, and specific humidity using the fundamental laws of physics (Gultepe *et al.*, 2007). These primary variables are then converted into visibility using parameterisation schemes (Tang *et al.*, 2009; van der Velde *et al.*, 2010; Zhou *et al.*, 2007). One popular example is the Weather Research and Forecasting (WRF) model using VIS-RH parameterisation scheme that estimates visibility using relative humidity (Lee *et al.*, 2021; Payra & Mohan, 2014; Pithani *et al.*, 2019; Pu *et al.*, 2016; Steeneveld *et al.*, 2016; Wagh *et al.*, 2023). Since the physics of fog formation is complex and sensitive to both local and large-scale climate processes, expecting a universally applicable parameterisation scheme is unreasonable (Long *et al.*, 2021). The parameterisation schemes are often calibrated using in situ hydro-meteorological measurements and have poor generalisation performance. A review of the past NWP-based fog forecasting methods is provided in Table S1 in Appendix 1, revealing that most studies have focused on threshold visibility of 500 m or less rather than the WMO-specified threshold of 1 km. As the lead time increases, the error component and corresponding computational cost also increase in NWP models (Singh *et al.*, 2025). Thus, medium-range fog forecasts for up to five days or longer are not yet widely studied.

Machine-learning weather prediction (MLWP) models, such as GraphCast, developed by Google DeepMind (Lam *et al.*, 2023), mark a significant advancement in weather forecasting. GraphCast employs graph neural networks (GNNs) to represent the Earth's atmosphere as a graph, with nodes and edges capturing spatial relationships. This approach enables the model to efficiently learn complex, nonlinear interactions in atmospheric data, offering improved accuracy and computational efficiency compared to traditional NWP models. GraphCast uses recent weather states from reanalysis datasets (e.g., ERA5; Rasp *et al.*, 2024) as inputs and forecasts future weather states, maintaining the same spatial-temporal resolution and variable set as the input data.

GraphCast's innovation lies in its ability to provide accurate forecasts of meteorological variables, such as temperature, pressure, and wind speed, over a 10-day horizon at six-hour intervals. Studies, such as the WeatherBench2 (Rasp *et al.*, 2024), demonstrate that GraphCast outperforms traditional forecasting systems, including the widely used IFS (Integrated Forecasting System; Rasp *et al.*, 2024). GraphCast is designed for downstream applications, such

as predicting floods, forecasting solar energy supply, agriculture planning, and aviation routing, which involve using weather forecasts as inputs for decision-making in other domains. However, the number of studies exploring such applications remains limited (Feldmann *et al.*, 2024; Flora & Potvin, 2025; Suri *et al.*, 2024; Yan *et al.*, 2024). While GraphCast is not explicitly designed for fog forecasting, its precise forecasts of fog covariates may be valuable for fog forecasting. Integrating global MLWP models like GraphCast with postprocessing techniques presents opportunities for developing fog forecasting models that have not yet been explored.

To address the limitations of current fog forecasting models, we introduce a hybrid system, termed ‘FogCast’, that leverages GraphCast forecasts and combines them with Bayesian neural networks (BNNs) to generate probabilistic fog forecasts. BNNs are uncertainty-aware neural networks capable of disentangling aleatoric uncertainty (reduces with more features) and epistemic uncertainty (reduces with a bigger dataset) (Murphy, 2022, Chapter 2).

FogCast is evaluated on the fog-prone region of IGP, which has been a testbed for developing fog forecasting models for research (Pithani *et al.*, 2020; Smith *et al.*, 2023) and operational applications (Arun *et al.*, 2022; NCMWRF, 2025; Sharma *et al.*, 2024). However, most fog forecasts are available only at airports in IGP with a maximum lead time of three days (Jayakumar *et al.*, 2021). Medium-range forecast models (with a lead time of five days) recently proposed by Bajaj *et al.* (2024) do not forecast fog directly but provide a fog index, which is a combined measure of fog duration and intensity. Further, these forecast models are site-specific. Thus, previous studies have not provided fog forecasts with lead times beyond five days or at a spatial scale encompassing the entire IGP, leaving a clear research gap.

The specific objectives of the study are:

1. To develop FogCast, the first medium-range (10-day lead time) fog forecasting model based on GraphCast.
2. To evaluate the model performance for different lead times and visibility-based fog types.
3. To disentangle the aleatoric and epistemic components of uncertainty in the fog forecasts.

This study distinguishes itself by evaluating FogCast on multiple winter seasons and across geographically diverse sites in the IGP, ensuring robustness and scalability. FogCast is trained and tested at selected representative lead times (6 h, 12 h, 1 day, 2 days, 5 days, and 10 days), rather than all 40 available six-hourly lead times, in order to balance computational feasibility with practical relevance.

2 | STUDY AREA AND DATASETS

The IGP is prone to persistent fog, particularly long spells of radiation fog during the winter months (November–February). The unique geography of the IGP—characterised by flat plains bordered by the Himalayan and Aravalli Mountain ranges—plays a critical role in its climatology (Deshpande *et al.*, 2023). Ample moisture from western disturbances, proximity to water bodies, and high aerosol concentrations further contribute to the persistence and severity of fog in the region. These factors create ideal conditions for the formation and maintenance of dense fog, significantly impacting visibility and exacerbating the region’s vulnerability to transportation and health disruptions.

The datasets used in this study are categorised into two types: in situ observational data (used as labels or reference data or ground truth) and weather forecast data (used as input features, i.e., input data). The in situ data comprise routine weather reports (METAR – Meteorological Aerodrome Reports) obtained from the Iowa Environmental Mesonet archive (accessible at: <https://mesonet.agron.iastate.edu/request/download.phtml>). METAR is a standardised international format for reporting current observations at airports widely used in meteorological and hydrological research. These METAR reports, typically issued every 30 minutes, include meteorological parameters such as visibility, temperature, wind speed, and relative humidity. Observations with visibility below 1 km are classified as fog observations in the reference dataset. For this study, ‘visibility data’ refers to general visibility, also known as meteorological visibility or meteorological optical range (Gultepe *et al.*, 2007).

The weather forecast dataset is derived from GraphCast and includes both surface-level variables and variables at different pressure levels. The surface-level variables consist of temperature at 2 m above ground, U and V components of wind at 10 m, and total precipitation over the last six hours. While the model is trained on data from selected cities, it is applied across the entire IGP, where the actual surface pressure depends on topography and prevailing conditions. Therefore, the 1000 hPa level is used as the most consistent and practical approximation of near-surface processes. The variables at 1000 hPa include temperature, specific humidity, and U , V , and W components of wind. Historical GraphCast forecasts (Lam *et al.*, 2023; Rasp *et al.*, 2024) are utilised for training and evaluating FogCast. For real-time fog forecasting, we use six-hourly experimental forecasts produced by NOAA’s Environmental Modelling Center, where the GraphCast architecture is fine-tuned with GDAS data (Tabas *et al.*, 2025). These forecasts are part

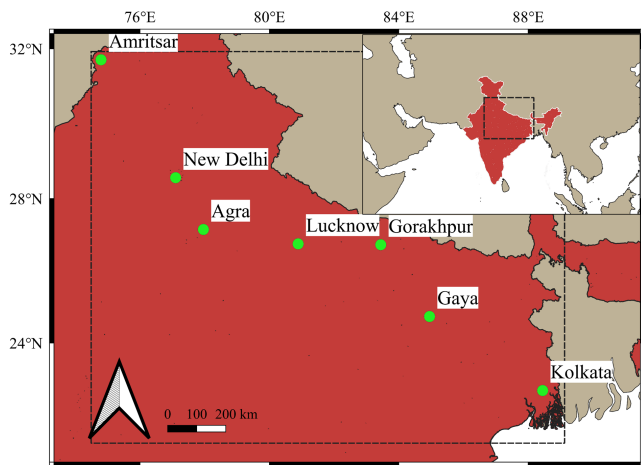


FIGURE 1 Locations of seven cities (small circles) in the Indo-Gangetic Plains used for model training and evaluation. The keymap in the upper right corner shows the study area (shown by dashed lines) at a coarser scale. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

of the Global-Eagle-Solo initiative and are distinct from NOAA's official operational products.

The study focuses on the winter season (November to February) from 2017 to 2023 for training and evaluating the model. GraphCast weather forecast data are extracted for grid points corresponding to seven cities (airports): Agra, Amritsar, Delhi, Gaya, Gorakhpur, Kolkata, and Lucknow (Figure 1). For each city, the visibility observation closest in time (within one hour) to the corresponding GraphCast forecast is obtained. These cities are selected because they have at least 25 fog observations (visibility <1 km) in the METAR dataset during the study period. To assess data completeness, Table S3 presents the total number of METAR observations and the fraction of fog observations for each city. It also shows the geographical setting of the cities, which is further discussed in Deshpande *et al.* (2023). The number of missing observations varies across cities, with Agra and Gaya each having approximately six months of completely missing METAR data. Model evaluation is performed only for periods with valid visibility observations. Moreover, since the FogCast model performance (e.g., Tables 2 and 3) is reported by aggregating data from all cities, the impact of unequal data availability across stations is minimised.

Fog, like many hydro-meteorological variables, exhibits temporal autocorrelation and diurnal variations. Therefore, lagged GraphCast forecasts up to the past 24 hours (four time-steps of six-hourly forecasts) are used as input features. In addition, the input features included the day of the corresponding winter season (DOS) and the hour of the day (Hour). To preserve the cyclic nature of DOS and Hour, both variables are converted to cosine

values. The resulting time-series of the input features, along with their lagged counterparts and corresponding binary labels (fog/no fog), are fed into FogCast for training and testing. A summary of the FogCast model details is provided in Table 1.

3 | METHODOLOGY

An overview of the methodology is provided in Figure 2, and its details are elaborated in this section.

3.1 | Theoretical background of Bayesian Neural Networks

In ordinary neural networks (also known as point estimate neural networks), weights (parameters) and the networks' output are deterministic quantities, while both weights and outputs of BNNs are random variables characterised by probability distribution functions. Suitable probability distributions are used as prior distributions of weights, which are updated during the model training phase. The posterior distributions of the weights represent the updated belief about the parameters after observing the training data. Since the exact posterior is analytically intractable, it is approximated as a normal (Gaussian) distribution using variational inference (Hüllermeier & Waegeman, 2021). The parameters of the normal distribution, that is, mean and variance, are optimized during training to minimise the divergence between the approximate and true posterior distributions. The details of the training procedure and hyperparameters involved in this model are provided in previous studies (Deshpande *et al.*, 2024; Kwon *et al.*, 2020).

The predictive distribution of the network output is estimated using Monte Carlo simulations, where weights are randomly sampled from their posterior distribution. The mean of these output samples provides the mean probability of fog forecast (Equation 1), and their standard deviation represents epistemic uncertainty (Equation 3).

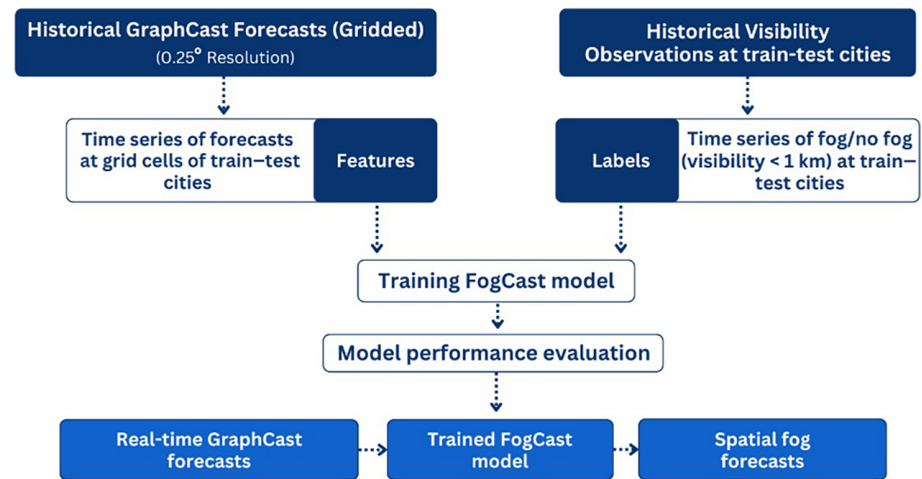
$$\bar{y} = \frac{1}{T} \sum_{t=1}^T \hat{y}_t, \quad (1)$$

where T is the number of Monte Carlo simulations and $\hat{y}_t \in [0, 1]$ is the fog probability for simulation t . In terms of standard deviation, aleatoric, epistemic, and total uncertainty are given by Kwon *et al.* (2020) and Hüllermeier and Waegeman (2021).

$$\text{Aleatoric uncertainty } (\sigma_a) = \sqrt{\frac{1}{T} \sum_{t=1}^T \hat{y}_t (1 - \hat{y}_t)}, \quad (2)$$

TABLE 1 Overview of the FogCast model and experimental setup.

Category	Details
Type of forecast	Binary classification (Fog/No Fog), probabilistic outputs using Bayesian Neural Networks (BNNs)
Region of interest	Indo-Gangetic Plains in India
Input features	- Surface: temperature at 2 m, U, V wind (10 m), total precipitation (six-hourly) 1000 hPa: temperature, specific humidity, U, V, and W wind - Four lagged features of six-hourly forecasts are used
Ground reference (labels)	METAR visibility observations (fog if visibility < 1 km)
Period of record	Winters (Nov–Feb), 2017–2023
Locations of model evaluations	Seven cities (airports) in the Indo-Gangetic Plains (Agra, Amritsar, Delhi, Gaya, Gorakhpur, Kolkata, Lucknow)
Dataset size	13,796 observations (~18% fog events)
Validation strategy	Leave-one-city-out cross validation (LOOCV) method
Forecast interval	Every six hours
Lead times evaluated	6 hours, 12 hours, 1 day, 2 days, 5 days, 10 days

FIGURE 2 Flowchart showing methodology of FogCast, a GraphCast-based fog forecasting model. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/qj.20150)]

$$\text{Epistemic uncertainty } (\sigma_e) = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{y}_t - \bar{y})^2}, \quad (3)$$

$$\text{Total uncertainty } (\sigma_u) = \sqrt{\sigma_a^2 + \sigma_e^2}, \quad (4)$$

since $\hat{y}_t$ are bounded between 0 and 1, the aleatoric and epistemic components are each limited to 0.5. For a given input, the aleatoric uncertainty is low when the mean prediction $\bar{y}$ is close to 0 or 1, reflecting low inherent randomness in the outcome. The epistemic uncertainty is low when the individual predictions $\hat{y}_t$ across stochastic forward passes are tightly clustered around $\bar{y}$, and increases when these predictions exhibit greater spread, reflecting higher uncertainty in the model parameters. Although Equation (4) defines the total uncertainty in terms of its

aleatoric and epistemic components, in this study, we analyse the two terms separately to highlight their distinct contributions.

3.2 | Model architecture

The BNN model is developed in a Python environment using the TensorFlow Probability (TFP) library (Dillon *et al.*, 2017). The first layer of the BNN model is a fully connected (TFP dense) layer with 120 neurons and uses the Rectified Linear Unit (ReLU) activation function (Figure 3). The ReLU activation function outputs zero for negative inputs and returns the input value for positive inputs (Fukushima, 1975), thereby introducing nonlinearity while avoiding issues such as vanishing gradients.

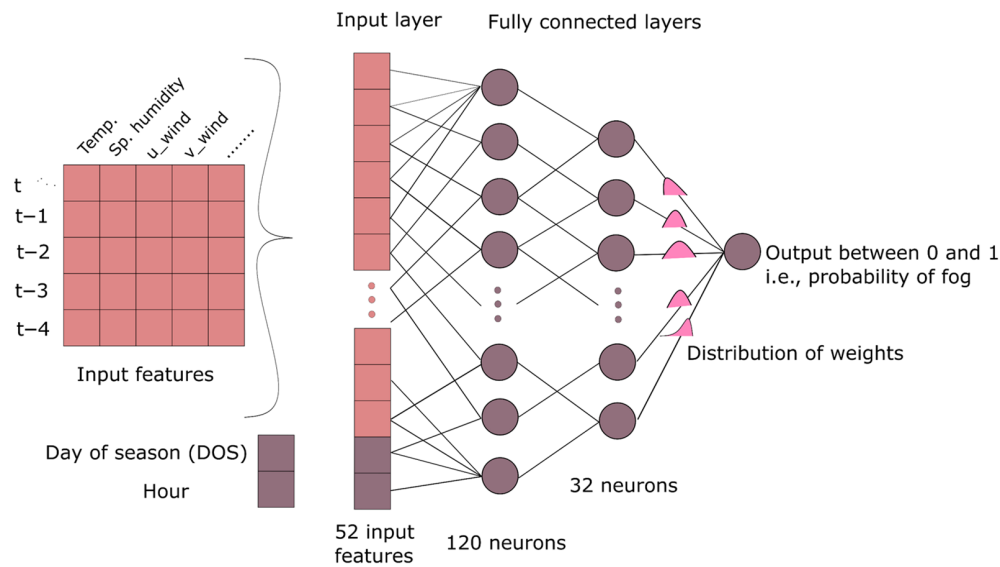


FIGURE 3 Bayesian neural network used in the FogCast. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/qj.20150)]

This pattern is repeated with a subsequent dense layer of 32 neurons. The third layer, a variational inference (TFP DenseVariational in the TensorFlow library) layer, is added to model epistemic uncertainty. The variational inference layer, is added to represent the model's parameters as probability distributions rather than fixed values. Unlike a standard dense layer, DenseVariational parameterises both the mean and variance of the weight distributions during training by employing variational inference (Chang, 2021). This approach allows the network to quantify epistemic uncertainty by propagating uncertainty in the model parameters through the model. The prior distribution of the weights in this layer is defined as a zero-mean, unit-variance multivariate normal distribution, since weights can theoretically take values from $-\infty$ to ∞ and normal priors are commonly used in Bayesian neural networks (Kendall & Gal, 2017). The final output layer employs a Bernoulli distribution to produce a value between 0 and 1, that is, the probability of fog. The network uses the negative log-likelihood of the Bernoulli distribution as its loss function, which is optimised using a stochastic gradient descent algorithm. Since the METAR dataset contains four times more fog observations than no-fog observations, a class imbalance exists. To compensate for the class imbalance, a weighted loss function is used that penalises the misclassification of fog observations approximately four times more than the misclassification of no-fog observations. The model is trained until the loss function converges or for a maximum of 200 epochs. For the trained model, 20 Monte Carlo simulations are performed, and the mean and standard deviation of the probability of fog forecasts are calculated to evaluate the results using Equations (2) and (3). The probability values thus obtained are converted into fog/no

fog based on a threshold of 0.5. Figure 3 shows the model architecture of the BNN used in FogCast.

3.3 | Training and testing

FogCast aims to forecast fog at locations where in situ visibility observations are unavailable. To mimic this situation, leave-one-out cross validation (LOOCV) is used. In each iteration, data from a single city are excluded from the training set and used as the test set, while data from the remaining cities form the training set. This process is repeated until every city has been used as the test set exactly once. The results are calculated as a weighted average (based on the number of fog observations) over all iterations. The model performance is evaluated using different metrics mentioned in Table 2. Due to class imbalance, Critical Success Index (CSI) is more important than Overall Accuracy (OA) as a decent OA can be obtained even by completely misclassifying the minority class (i.e., fog). Similarly, Probability of Detection (PoD), that is, recall, is more important than precision in fog forecasting.

3.4 | Feature importance analysis

To understand the impact of each input feature on the FogCast forecasts, an ablation study is carried out using SHapley Additive exPlanations (SHAP) values (Lundberg & Lee, 2017). SHAP values provide a unified measure of feature importance by attributing the impact of each input feature on fog forecast. For each feature, the SHAP value calculation considers different combinations of subsets of features for estimating the feature importance. The

TABLE 2 Model performance for fog forecasting averaged over all lead times during testing (visibility threshold:1000 m).

Metric	Range	Ideal value	Formula	Lead time	
				6 hours	10 days
Critical Success Index (CSI)	[0,1]	1	$\frac{TP}{TP+FP+FN}$	0.440	0.362
Equitable Threat Score (ETS)	$[-1/3,1]$	1	$\frac{TP - TP_{\text{random}}}{TP+FP+FN}$	0.304	0.199
Overall Accuracy (OA)	[0,1]	1	$\frac{TP+TN}{TP+FP+FN+TN}$	0.776	0.684
Probability of Detection, i.e., PoD (Recall)	[0,1]	1	$\frac{TP}{TP+FN}$	0.766	0.712
Precision	[0,1]	1	$\frac{TP}{TP+FP}$	0.517	0.444
False Alarm Rate (FAR)	[0,1]	0	$\frac{FP}{TN+FP}$	0.228	0.328
Hanssen–Kuiper Skill Score	$[-1,1]$	1	PoD – FAR	0.538	0.384

Note: TP, correctly forecasted fog observation (hit); TN, correctly forecasted no fog (correct negative); FP, incorrectly forecasted no fog (false alarm); FN, incorrectly forecasted fog (miss). $TP_{\text{random}} = \frac{(TP+FP)(TP+FN)}{TP+FP+FN+TN}$.

FogCast forecast is computed for each subset. A SHAP summary plot is obtained across all lead times and all test cities together to get an overall idea of the importance of features.

4 | RESULTS AND DISCUSSION

4.1 | Model evaluation

FogCast forecasts fog (visibility <1000 m) with a lead time of up to 10 days. However, for model evaluation lead times of 6, 12 hours and 1, 2, 5, 10 days are selected and the performance for the shortest and the longest lead times is presented in Table 2 and Figure 4. As expected, the model performance is generally better for shorter lead times. However, one of the most striking results is that the model's performance (in terms of CSI) at 10 days lead time is within 20% of the performance at six hours lead time. A similar trend is also observed in Equitable Threat Score (ETS) and Hanssen–Kuiper Skill Score. This capability of FogCast to forecast fog for longer lead times without significant performance deterioration is likely a consequence of GraphCast's ability to forecast meteorological variables at longer lead times (Lam *et al.*, 2023). FogCast performance is compared with both statistical (Kim *et al.*, 2024; Sharma *et al.*, 2024) and NWP-based fog forecasting studies in IGP (Parde *et al.*, 2022; Payra & Mohan, 2014; Singh *et al.*, 2022) using the performance metrics reported in those works, rather than through a common dataset. The CSI values of the existing studies vary from 0.27 to 0.65, whereas the PoD and False Alarm Rate (FAR) values vary from 0.60 to 0.95 and 0.04 to 0.43, respectively (detailed results are provided in Table S1 in Supporting Information). However, all past models have different visibility thresholds, spatiotemporal resolutions, and testing locations, with the longest lead time up to two days (except one study with five days lead

time), making one-to-one comparison difficult, highlighting the need for a standardised dataset for fog forecasting. Nevertheless, FogCast shows comparable performance (in terms of CSI, PoD, and FAR) to these past studies, even with a lead time that is more than twice as long as theirs.

Figure 4 shows that both aleatoric and epistemic uncertainty (median values) increase as the lead time increases. Thus, as expected, the model is relatively more confident in forecasting a shorter lead time, and it becomes more and more uncertain as the lead time increases. A detailed discussion on uncertainty analysis is provided in a later subsection.

4.2 | Model evaluation for visibility-based fog categories

In the previous subsection, the model results are provided for the entire visibility range; however, it is important to understand the model performance for different visibility-based fog categories as they have varying impacts. The results for different fog categories are presented in Table 3. The column “Fog (%)” indicates the proportion of samples that fall within each category. In other words, it shows the percentage of observations belonging to the corresponding category. The model shows higher performance for dense and very dense fog categories (PoD/CSI of 0.86 and 0.93, respectively), indicating that the FogCast confidently forecasts these fog categories as compared to shallow fog (PoD/CSI of 0.62). A possible reason for the relatively higher misclassification of shallow-fog observations could be the presence of hygroscopic aerosols (that attract moisture and facilitate droplet growth) and non-hygroscopic aerosols (that contribute to haze by scattering the light), which reduce visibility without the formation of fog droplets (Deying *et al.*, 2021). Additionally, visibility is measured at a point scale, while

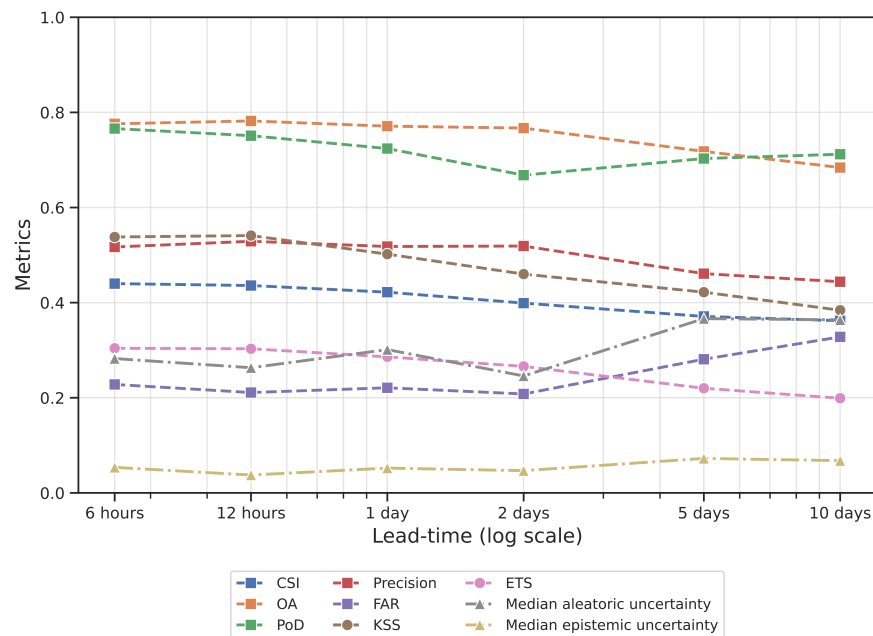


FIGURE 4 FogCast performance for different lead times across all seven cities during testing. The formulae for the metrics used are provided in Table 2, and the threshold visibility is 1000 m. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/qj.20150)]

TABLE 3 Model performance for different visibility-based fog categories across all lead times and all seven cities during testing.

Visibility range (m)	Fog category	Fog (%)	PoD		Median aleatoric uncertainty		Median epistemic uncertainty	
			6 hours	10 days	6 hours	10 days	6 hours	10 days
0–50	Very dense	2.52	0.95	0.93	0.28	0.33	0.06	0.06
50–200	Dense	1.62	0.92	0.86	0.28	0.37	0.05	0.06
200–500	Medium	3.15	0.83	0.76	0.30	0.39	0.06	0.06
500–1000	Shallow	10.52	0.68	0.62	0.35	0.39	0.07	0.07
>1000	No fog	82.19	0.83	0.68	0.20	0.35	0.04	0.06

Note: The Fog (%) column corresponds to the percentage of samples that have fallen into a specific fog category. The metrics are calculated for subsets of the test data selected based on visibility ranges; hence, there is no complete confusion matrix across all categories. In such cases, the Probability of Detection (PoD) effectively becomes equivalent to the Overall Accuracy (OA) and the Critical Success Index (CSI).

the GraphCast variables used as input features have a spatial resolution of approximately 25 km. Lastly, dense fog events are usually sustained and widespread, increasing the likelihood that the entire grid cell is covered with fog, which improves consistency between the large-scale predictors and point-scale observations. Such events are also easier for the model to learn, as they produce clearer distinctions in the predictor space. In contrast, shallow-fog events are more localised, making them harder to capture given the resolution mismatch. However, since the proportion of shallow fog is higher as compared to dense and very dense fog, the model's overall performance is dominated by its performance in the shallow-fog category.

4.3 | Uncertainty analysis

Figure 5 displays the distribution of aleatoric and epistemic uncertainties (shown as box plots on the

margins) along with their scatterplot, and Table 4 presents the corresponding statistics. The analysis reveals that aleatoric uncertainty is generally higher than epistemic uncertainty, with mean values of 0.27 and 0.06, respectively, suggesting sufficient training data but insufficient features to separate fog/no-fog classes. Low epistemic uncertainty suggests the model was trained on sufficient, representative data and that the features are likely to capture key mechanisms. In contrast, high aleatoric uncertainty reflects inherent noise or ambiguity in the data, which cannot be reduced simply by adding more data. This could be due to unobserved variables or natural variability in fog formation, such as microclimatic effects, for example, localised humidity and cooling caused by small water bodies or irrigated agricultural fields, which are often not captured in coarse-resolution datasets. Also, the imbalanced dataset, with four times more no-fog than fog cases, can influence uncertainty. Even with good

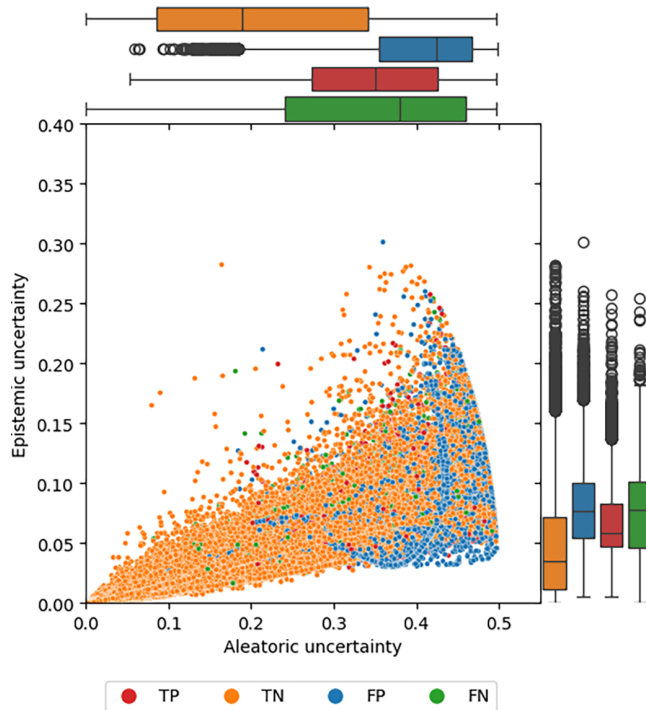


FIGURE 5 Scatterplot for aleatoric and epistemic uncertainty. The boxplots for aleatoric and epistemic uncertainties are shown on the top and right, respectively. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

TABLE 4 Statistics of aleatoric and epistemic uncertainties for binary fog vs no-fog classification.

	Aleatoric				Epistemic			
	TP	FP	TN	FN	TP	FP	TN	FN
Mean	0.34	0.40	0.22	0.34	0.07	0.08	0.05	0.07
Median	0.35	0.42	0.19	0.38	0.06	0.08	0.04	0.08
CV	0.28	0.20	0.69	0.40	0.45	0.41	0.87	0.53
Mean	0.27				0.06			
Median	0.28				0.05			
CV	0.56				0.67			

Note: TP, correctly forecasted fog observation (hit); TN, correctly forecasted no fog (correct negative); FP, incorrectly forecasted no fog (false alarm); FN, incorrectly forecasted fog (miss). Reported values include the mean, median, and coefficient of variation (CV, standard deviation/mean). The first three rows present results for each element of the confusion matrix (TP, FP, TN, FN) separately, while the last three rows summarise results combined across all four categories.

features, some outcomes may remain unpredictable due to real-world variability.

The coefficient of variation for aleatoric uncertainty (0.56) is lower than for epistemic uncertainty (0.67), indicating that, though the aleatoric uncertainty may take higher values, epistemic uncertainty has larger

variation. The scatterplot suggests that forecasts with higher aleatoric uncertainty often have higher epistemic uncertainty. The aleatoric and epistemic uncertainties are conceptually distinct but appear to be correlated to the fog forecast, as the observations far away from the transition zone (dense, very dense, and no-fog observations) are relatively higher in number and contain informative features leading to lower epistemic and aleatoric uncertainties. Table 4 and boxplots of Figure 5 highlight that correct forecasts (true positives, TPs, and true negatives, TNs) have lower median aleatoric and epistemic uncertainties than incorrect forecasts (false positives, FPs, and false negatives, FNs). Thus, FogCast is more confident in correct forecasts than incorrect forecasts.

4.4 | Feature importance

The feature importance estimated across all lead times and all test cities together is plotted as a SHAP value plot in Figure 6a. The position on the x-axis shows the SHAP values and the y-axis shows the input features such as temperature, humidity, or wind speed. The top features have a higher average impact on the forecast. The temperature 2 m above ground is found to be the most influential feature, followed by the specific humidity at 1000 hPa. These two variables control the relative humidity that plays a crucial role in fog formation. The plot also shows that the temperature at 2 m above ground has higher importance than the temperature at 1000 hPa, indicating that the fog formation is more influenced by the features at the surface level. Wind components, along with mean sea level pressure, are identified as the most influencing factor after temperature and specific humidity.

The Hour variable representing the time of day is also found to be important, that is, the time of the day also affects the chances of fog. This inference is aligned with fog climatology in IGP that suggests a strong diurnal pattern in fog occurrence with most fog observed during dawn, dusk, and night-time (Deshpande *et al.*, 2023). The rest of the input features have low SHAP values, indicating an insignificant contribution to fog forecasting. Latitude, longitude, and elevation of cities were initially included as input features. However, since the current model is trained using only seven locations, these variables take on just seven distinct values of latitude–longitude pairs. This limited variability restricts the model's ability to learn its broader influence, and consequently, the SHAP analysis assigns them very small values. Therefore, these features are not retained in the final FogCast model. Though widespread fog episodes exhibit spatial continuity, the limited number of cities under study limits the learning of these spatial connections.

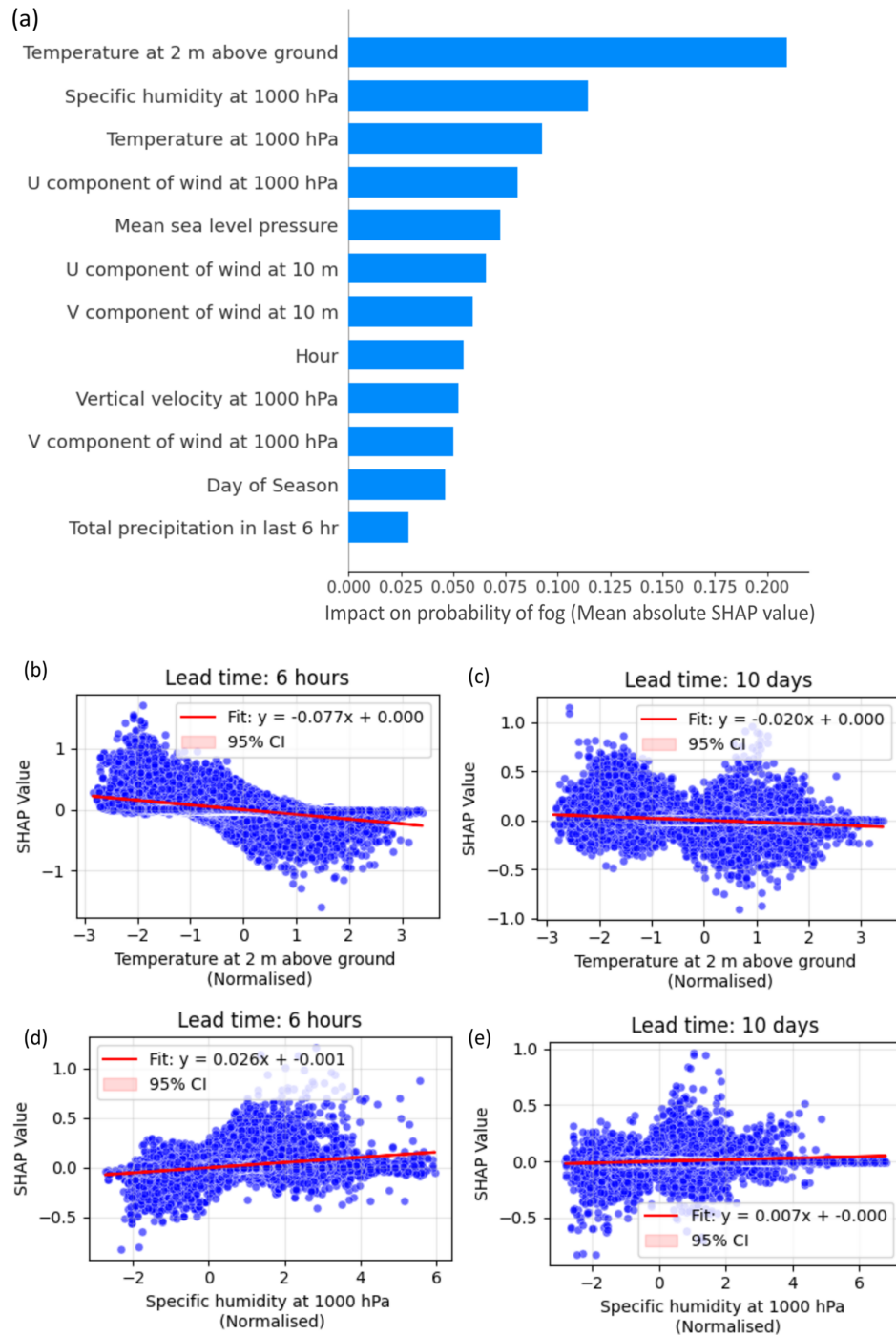


FIGURE 6 (a) Feature importance plot showing the input features in descending order of feature importance. (b–e) Scatterplots for normalised feature value and corresponding SHAP value. Positive SHAP values (shown on the y-axis) represent higher fog probability and vice versa. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

To study the effect of temperature at 2 m above the ground and specific humidity at 1000 hPa on fog forecasting for individual observations at different lead times, scatterplots are drawn between SHAP values and these features (Figure 6b–e). For lower values of temperature and higher values of specific humidity, the SHAP values are positive, suggesting a higher probability of fog. This pattern is clearly visible in the lead time of six hours (Figure 6b,d) but not with a lead time of 10 days (Figure 6c,e), indicating that while forecast skill remains reasonable at longer lead times, the

physical interpretability of feature contributions weakens and should be treated with caution.

4.5 | FogCast performance in extreme events: Two case studies

The results shown in the previous section provide the quantitative performance of the model, however, a popular way to assess the weather forecast model performance is to study its performance for an extreme weather event

TABLE 5 Model performance for a severe fog event in Delhi (India) on 14 January 2022. (Times shown are in Indian Standard Time, i.e., IST, UTC + 0530 h).

Date of forecast	Lead time	14 January 2022		
		0:30	6:30	12:30
13-Jan-22	6 hours	✓	✓	✓
13-Jan-22	12 hours	✓	✓	✓
13-Jan-22	1 day	✓	✓	✓
12-Jan-22	2 days	✓	✓	X
9-Jan-22	5 days	✓	✓	X
4-Jan-22	10 days	✓	✓	✓
Correct forecasts out of 6		6	6	4

Note: ✓, correct forecast; X, incorrect forecast.

(Price *et al.*, 2024). The subsequent paragraphs present FogCast performance for two fog events with longer duration (>12 hours) and thicker fog appearance, one over New Delhi on 14 January 2022, and another over many parts of IGP on 4 January 2025.

Western disturbances are one of the major causes of widespread dense fog events in IGPs by providing large amounts of moisture content. One such western disturbance was observed near Delhi on 13 January 2022 (Hunt *et al.*, 2025) followed by a fog event in Delhi. In this study, the model performance is studied for a severe fog event that happened in Delhi on 14 January 2022. It was the longest reported fog event in which visibility was less than 300 m for most of the event. This fog event caused severe disruptions in transportation services, including flight delays (TIE, 2022; TOI, 2022). The FogCast could identify this fog event with even 10 days lead time (Table 5). For lead times of six hours to 10 days, the corresponding mean aleatoric uncertainty values vary from 0.34 to 0.41, whereas median epistemic uncertainty values vary from 0.04 to 0.06. However, the relatively small difference in epistemic uncertainty values between six-hour and 10-day forecasts indicates that the model retains a stable level of confidence in its learned parameters across different lead times. This stability is likely because models for all lead times are trained on a similar quantity of data, which imparts comparable confidence in the learned parameters.

To study FogCast's forecasting ability in a real-time operational setting, a recent fog event is selected. Figure 7a shows the probability of fog forecasted by FogCast with a one-day lead time for 4 January 2025 at 0530 h (UTC + 0530), while Figure 7b displays the corresponding fog detected by INSAT-3D Fog, an operational satellite fog-monitoring product developed by the Indian Space Research Organization (ISRO) that provides the

near-real-time fog updates for India and neighbouring countries. The INSAT-3D Fog product, derived from the geostationary multispectral INSAT-3D satellite (Indian Space Research Organisation, 2015), provides fog detection at a spatial resolution of 4 km and a 30-minute temporal interval. The product identifies fog using the brightness temperature difference between the 3.9 μm and 10.8 μm thermal infrared channels (Chaurasia & Jenamani, 2017). However, INSAT-3D Fog shows a limited performance of CSI and PoD, 0.19 and 0.31, respectively, over IGP (Deshpande *et al.*, 2022). Thus, the FogCast forecast is also compared (Figure 7c) with the INSAT-BNN fog detection method (Deshpande *et al.*, 2024) that uses various bands of INSAT-3D images and shows relatively better performance (CSI and PoD, 0.48 and 0.68, respectively). The area under the fog appears largely similar in all three images, suggesting that the fog forecasted by FogCast agrees with the fog detected by INSAT-3D satellite. Table 6 compares the FogCast forecast with INSAT-3D Fog, INSAT-BNN fog, and observations at METAR sites for this fog event. The FogCast forecasts closely align with METAR observations, particularly in Amritsar, Delhi, and Lucknow, where all three sources agree on fog occurrence. In Agra and Gaya (and Kolkata), both FogCast and INSAT-3D Fog show agreement on fog (and no fog); however, METAR observations are unavailable for these cities. In New Delhi, METAR, INSAT-BNN, and FogCast report the presence of fog, but INSAT-3D Fog reports no fog. Lastly, in Gorakhpur, FogCast (and INSAT-BNN Fog) and INSAT-3D Fog show conflicting results. Since the corresponding METAR observation is missing, it is unclear which one is more accurate. Nevertheless, FogCast performs comparably to METAR observations, INSAT-BNN Fog, and INSAT-3D Fog, both qualitatively and quantitatively.

4.6 | Further discussion

To assess the contribution of the lagged features to model performance, FogCast is trained and tested without them. This variant of FogCast without lagged features is termed as 'FogCast-lite'. The comparison of model performance with and without lagged features shows a modest decrease in CSI (and in PoD) and an increase in uncertainties (especially aleatoric uncertainty). Thus, the temporal autocorrelation contributes to the model performance, though its importance decreases with the lead time (Table S2 in Supporting Information).

The details of the model performance for each city in the LOOCV method reveal that the CSI values vary significantly. In the case of Kolkata and Gaya (0.10 and 0.22, respectively), the model performance is inferior compared to that for the remaining cities (Table S3 in Supporting

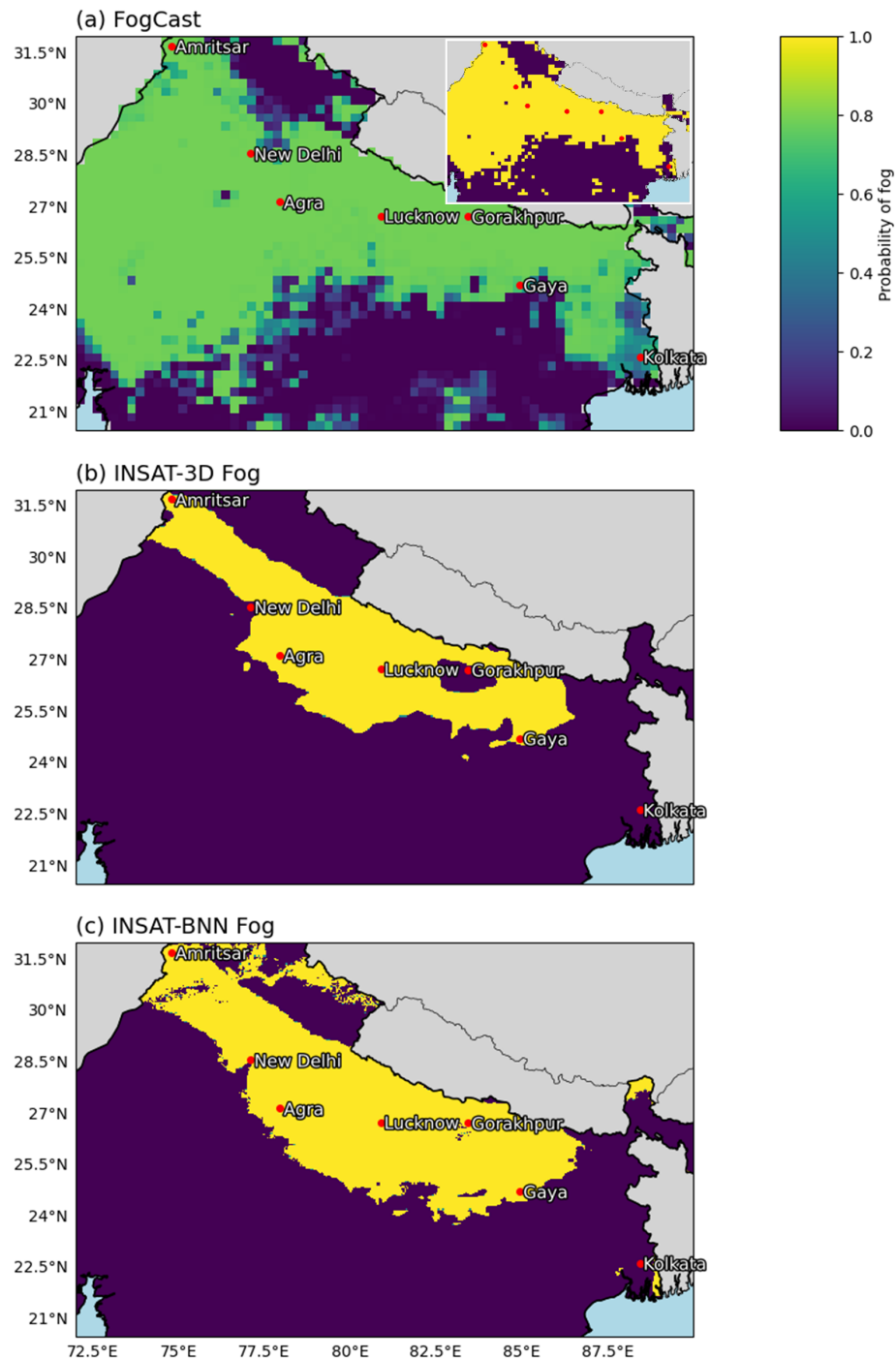


FIGURE 7 Fog conditions on 4 January 2025 at 0530 h. (a) Probability of fog forecasted by FogCast with a one-day lead time; the inset shows the binary fog forecast (using a probability threshold of 0.5). (b) Fog detected using the operational INSAT-3D fog product. (c) Fog detected using the INSAT-BNN fog detection method (Deshpande *et al.*, 2024). Yellow colour indicates fog. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/qj.12039)]

Information). The primary objective of the FogCast is to provide fog forecasts across the entire IGP rather than city-specific predictions. Thus, to avoid bias, the relatively low-performing cities are retained. The nature of fog in Kolkata is different from that in other cities in IGP due to its proximity to the coast (Deshpande *et al.*, 2023), posing a limitation on the model performance. In the case of Gaya, the dataset contains only 29 fog observations (~3% of all observed data in Gaya) with no dense and very dense fog observations. Since FogCast performance for medium and shallow fog is relatively lower compared to dense and very

dense fog (Table 3), it is reflected in lower values of CSI and PoD but higher values of OA for Gaya. On the other hand, Lucknow shows relatively higher performance, as it is centrally located in the IGP and is surrounded by flat plains. However, the reasons discussed above for the variations in the performance of different cities are only indicative, and further investigation is necessary before drawing conclusions.

In the FogCast model, the default probability threshold is kept at 0.5 to derive binary fog forecasts used in the performance evaluation. However, the users may directly use

TABLE 6 Comparison of fog forecast from FogCast, fog detection from INSAT-3D Fog, INSAT-BNN, and fog observations from METAR on 4 January 2025, at 0530 h, based on Figure 7.

City	METAR	FogCast	INSAT-3D fog	INSAT-BNN
Amritsar	Fog	Fog	Fog	Fog
New Delhi	Fog	Fog	No fog	Fog
Agra	No data	Fog	Fog	Fog
Lucknow	Fog	Fog	Fog	Fog
Gorakhpur	No data	Fog	No fog	Fog
Gaya	No data	Fog	Fog	Fog
Kolkata	No data	No fog	No fog	No Fog

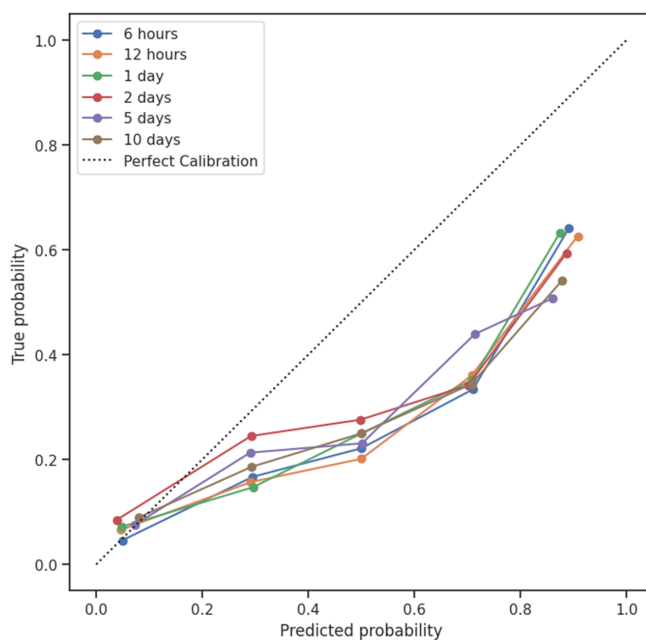


FIGURE 8 Reliability diagram showing the calibration of the FogCast for different lead times in different colours. The dotted line represents the perfectly calibrated classifier. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

the probability of fog rather than the binary fog forecasts. To compare the forecast probabilities and actual probabilities, a reliability diagram (Deshpande *et al.*, 2024; Guo *et al.*, 2017) is plotted (Figure 8). The *x*-axis of the reliability diagram represents the probability of fog estimated by the model, while the *y*-axis represents the fraction of correct forecasts (i.e., accuracy). The test dataset is sorted in ascending order of forecast probabilities and divided into bins. For each bin, the accuracy is plotted against the mean probability to get the calibration curve. For an ideal model, the calibration curve lies on a 45° line (shown as a black dotted line in Figure 8). The calibration curve for FogCast shows a deviation from the ideal curve around a

probability of 0.5, indicating an underprediction. Earlier results (Table 3) show that the dense and very dense fog observations with fog probability near to 1 are forecasted with a higher accuracy. Similarly, no-fog observations with fog probability near 0 are also forecasted with higher accuracy. Thus, the extreme probability values of the reliability diagram are better calibrated (showing lesser deviation) than the probability values around 0.5, which represent light and medium fog.

5 | SUMMARY AND CONCLUSION

FogCast, a Bayesian neural network-based probabilistic fog forecasting model, is developed utilising the GraphCast weather forecasting system. FogCast is tested for multiple winter seasons (2017–2023) across seven cities in the IGP. The model forecasts fog at six-hourly intervals with lead times ranging from six hours to 10 days. The CSI of the model varies from 0.44 for a six-hour lead time to 0.36 for a 10-day lead time. The lead time of FogCast is more than twice that of the existing methods, making it a novel tool for medium-range fog forecasting. The higher CSI values, even at longer lead times, underscore the model's robustness in maintaining reliable forecasts, making it particularly valuable for operational medium-range forecasts. The model shows high accuracy in detecting very dense fog (visibility <50 m), achieving a PoD exceeding 90%. The aleatoric uncertainty in fog forecasts is generally higher than the epistemic uncertainty, suggesting that more informative features are required to improve the model performance. It is observed that the temporal autocorrelation, captured through lagged features, improves model performance. Model performance varies across cities, reflecting differences in fog regimes and data availability; nevertheless, the model is designed to provide regionally consistent forecasts across the IGP rather than city-specific optimisation. Reliability analysis indicates that probabilistic forecasts are better calibrated for extreme fog and no-fog conditions than for intermediate probabilities, suggesting greater uncertainty in forecasting light and medium fog.

This study has several limitations, from which future research directions emerge. Forecast errors arise from errors in reanalysis data used by GraphCast, errors propagated from GraphCast forecasts themselves, and the inherent difficulty of representing localised fog using coarse-resolution inputs. While FogCast currently adopts a binary classification framework, extending it to multi-class fog forecasts or regression-based visibility forecasts could provide more detailed information, though such approaches remain challenging due to severe class imbalance. Future work may also explore the use of additional GraphCast pressure levels, higher temporal-resolution

forecasting, and city-specific models that combine local observations with large-scale forecasts. Incorporating large-scale climate drivers such as cold waves and western disturbances, as well as leveraging emerging foundation models (Rasul *et al.*, 2023) and spatially aware deep-learning approaches, offers further scope for improvement.

Despite these limitations, FogCast demonstrates the feasibility and value of data-driven, uncertainty-aware medium-range fog forecasting. The methodology is flexible and can be adapted to other machine-learning-based weather prediction systems such as GenCast (Price *et al.*, 2024), FuXi (Chen *et al.*, 2023), and FourCast-Net (Kurth *et al.*, 2023). Real-time FogCast forecasts are publicly available at https://fog.iitk.ac.in/fog-prediction/fogcast_lite_forecast.php, providing actionable information for transportation, agriculture, and public safety planning across the IGP.

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DATA AVAILABILITY STATEMENT

The publicly available datasets used in this research, viz., Iowa Environmental Mesonet (IEM) archive

(<https://mesonet.agron.iastate.edu/request/download.phtml>), National Oceanic and Atmospheric Administration (NOAA) (<https://noaa-nws-graphcastgfs-pds.s3.amazonaws.com/index.html#graphcastgfs>) are cited in the manuscripts at suitable locations.

ORCID

Prasad Deshpande  <https://orcid.org/0000-0001-6389-533X>

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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